

Lipschitz dependence of the limit coefficient on phase and amplitude

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Abstract

Unit 210 (programme A2, step 2) is the limit-side companion of unit 209. The functional $(\xi, \eta) \mapsto \text{phaseCoeff}(\xi, \eta) = D \int \eta(\pi u) J_p(\xi(\pi u)) w(u) du$ of Headline XIII is Lipschitz in sup norm on bounded input sets:

$$|\text{phaseCoeff}(\xi, \eta) - \text{phaseCoeff}(\xi', \eta')| \leq D(\|\eta - \eta'\|_{\infty} J_p(M) + \beta A \|\xi - \xi'\|_{\infty} J_{p+1}(M)) \int w$$

for $|\xi|, |\xi'| \leq M$, $|\eta'| \leq A$ on the closed cube `(phaseCoeff_sub_le)`, where $J_{p+1}(M) = \int_0^\infty s^p e^{-\beta s^2 + \beta M s} ds$. The moment-level inputs are monotonicity $J_p(a) \leq J_p(M)$ for $a \leq M$ and $|J_p(a) - J_p(a')| \leq \beta |a - a'| J_{p+1}(M)$, both integrated from the kernel bounds of unit 209. With `normalised_lipschitz_eventually` this makes the whole family $\{F_N\}_{N \geq N_0} \cup \{F\}$ equi-Lipschitz on bounded input sets, the hypothesis of the uniform-convergence step. Zero sorry/axiom; 9087 jobs.

1 The things formalised

```
theorem phaseMoment_le_of_le (h) (hp : 0 < p) (haM : a ≤ M) : phaseMoment p a
  phaseMoment p M
theorem phaseMoment_sub_le (h) (hp) (ha : |a| ≤ M) (ha' : |a'| ≤ M) :
  |phaseMoment p a - phaseMoment p a'| ≤ |a - a'| * phaseMoment (p + 1)
  M
theorem phaseCoeff_sub_le (hk) (hl) (h) (hmin) (continuity) (| |, | '| ≤ M) (| '|
  A)
(| - '| ≤ ) (| - '| ≤ on closedCube d) :
|phaseCoeff h k l - phaseCoeff h k l - '| ≤
  2 ^ (multCount (ratioExp h k) l - 1) * 2 * faceNorm h k l
  * (( * phaseMoment (2 l) M + * A * * phaseMoment (2 l + 1) M)
    * u in unitBox d, residualWeight h k l u)
```

2 Mathematics

Pointwise on the face, $\eta J(\xi) - \eta' J(\xi') = (\eta - \eta') J(\xi) + \eta' (J(\xi) - J(\xi'))$ with $J(\xi) \leq J(M)$ and $|J(\xi) - J(\xi')| \leq \beta |\xi - \xi'| J_{p+1}(M)$; the residual weight is nonnegative, so the bound integrates. $s^{p-1} \cdot s = s^{(p+1)-1}$ turns the extra factor s from the kernel bound into the moment of order $p + 1$ (`Real.rpow_add_one`).

3 Lean notes

- `gcongr` leaves the side goal $0 \leq J_{p+1}(M)$ when the atom is an opaque integral; discharge it with `all_goals first | exact ... | exact (phaseMoment_pos ...).le` rather than guessing the goal order.
- `set v := faceProj h k l u` folds every occurrence and keeps the pointwise calculation readable; the bounds `h M v h` apply because `faceProj_mapsTo` lands in the closed cube.

4 Next

Unit 211: inputs as elements of $C([0, 1]^d, \mathbb{R})^2$ via the clamp extension $u \mapsto (\max(0, \min(1, u_i)))_i$; equi-Lipschitz $\Rightarrow$ equicontinuous on bounded sets; pointwise convergence $\Rightarrow$ uniform convergence on compact input sets. Unit 212: random inputs, using tightness of the limit law in the Polish space $C([0, 1]^d)^2$ and the small-probability approximation lemma.